Companion XXI

Cosmic Reionization and the Thermal History of the Universe in the Relaxation-Driven Cyclic Cosmology

Ionizing Sources, IGM Heating, Reionization Morphology, and RDCC Photon–Baryon Coupling

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Abstract

The epoch of reionization (EoR) marks the transition from a neutral to an ionized Universe and encodes essential information about early structure formation, the first stars and galaxies, and the thermal history of the intergalactic medium (IGM). In the standard ΛCDM model, reionization is driven by early star formation and regulated by feedback, radiative transfer, and the growth of small-scale structure. In the Relaxation-Driven Cyclic Cosmology (RDCC), however, the relaxation-induced modification of gravitational dynamics directly affects the formation of early halos, the production of ionizing photons, and the thermal evolution of the IGM.

In this Companion we develop the first RDCC reionization framework. We derive how relaxation-modified collapse thresholds, scale-dependent growth, and baryonic suppression alter the abundance of ionizing sources, the escape fraction of UV photons, and the heating and ionization history of the IGM. We show that RDCC predicts a delayed onset of reionization, a smoother ionization morphology, reduced small-scale clumping, and a distinct 21-cm signal with a characteristic transition scale tied to the relaxation scale $k_{\rm rel}$.

Building on the baryonic physics developed in Companion XX and the non-linear structure formation analysis of Companion XIX, we construct an RDCC-adjusted reionization model and derive its impact on the ionization fraction, optical depth, IGM temperature, and 21-cm brightness temperature. The resulting predictions are consistent with current constraints from Planck, HERA, LOFAR, and JWST, and provide clear targets for upcoming surveys such as SKA and the Roman Space Telescope.

This Companion establishes cosmic reionization as a powerful observational probe of the relaxation dynamics of RDCC and extends the framework into the earliest luminous epoch of the Universe.

1 Motivation and Overview

The epoch of reionization (EoR) marks the transition of the Universe from a neutral to an ionized state and represents one of the most informative windows into early structure formation. The timing, morphology, and thermal evolution of reionization encode the abundance of early galaxies, the efficiency of ionizing photon production, the strength of feedback, and the thermodynamic state of the intergalactic medium (IGM).

In the standard ΛCDM model, reionization is driven by the first generations of stars and galaxies, with the ionization history determined by the growth of small-scale structure and the escape fraction of UV photons. However, several persistent tensions challenge this picture:

- the unexpectedly low stellar masses of early galaxies observed by JWST,
- the low Thomson optical depth inferred by Planck,
- the slow thermal evolution of the IGM at $z \sim 5$–6,
- and the absence of strong 21-cm absorption features in current HERA/LOFAR data.

These observations suggest that early structure formation and the thermal history of the IGM may differ from the standard scenario.

1.1 RDCC Perspective

The Relaxation-Driven Cyclic Cosmology (RDCC) provides a natural mechanism for modifying early structure formation. As shown in Companion XVIII and XIX, the relaxation rate $\Gamma(a)$ introduces a new physical scale that suppresses small-scale power, delays halo collapse, and reduces early star formation. These effects directly influence reionization:

- **Delayed formation of ionizing sources:** Fewer low-mass halos form at early times, reducing the early UV photon budget.
- **Reduced star formation efficiency:** The baryonic suppression derived in Companion XX lowers the ionizing emissivity.
- **Modified escape fractions:** Shallower potentials and reduced gas densities increase the escape fraction of UV photons.
- **Altered IGM heating:** Reduced small-scale structure lowers shock heating, while relaxation-induced vorticity enhances turbulent heating.
- **Smoother ionization morphology:** The suppression of small-scale clustering leads to larger, more coherent ionized regions.

Thus, reionization in RDCC is not merely a shifted version of the ΛCDM scenario but a distinct physical process shaped by relaxation-modified gravitational dynamics.

1.2 Goals of This Companion

The goals of this Companion are:

- to derive the RDCC-modified ionizing emissivity and escape fraction,
- to compute the RDCC reionization history and optical depth,
- to model the RDCC thermal evolution of the IGM,
- to analyze the morphology of ionized regions in the relaxation-modified regime,
- and to predict the RDCC 21-cm brightness temperature and power spectrum.

These results build directly on the baryonic physics developed in Companion XX and the non-linear structure formation analysis of Companion XIX.

1.3 Structure of This Companion

The structure of this Companion is as follows:

- Section 2 derives the RDCC-modified ionizing emissivity and escape fraction.
- Section 3 computes the RDCC reionization history and optical depth.
- Section 4 analyzes the thermal evolution of the IGM in RDCC.
- Section 5 examines the morphology of ionized regions and the 21-cm signal.
- Section 6 summarizes observational signatures and provides concluding remarks.

Together, these results establish cosmic reionization as a powerful observational probe of the relaxation dynamics of RDCC.

2 RDCC-Modified Ionizing Emissivity and Escape Fraction

The ionizing emissivity and the escape fraction of UV photons are the two key ingredients that determine the timing and morphology of cosmic reionization. In the standard ΛCDM framework, both quantities are governed by the abundance of low-mass halos, the star formation efficiency, and the structure of the interstellar medium (ISM). In RDCC, the relaxation-modified gravitational dynamics alter each of these components.

This section derives the RDCC-modified ionizing emissivity, escape fraction, and photon production efficiency.

2.1 Ionizing Emissivity in RDCC

The comoving ionizing emissivity is defined as

$$\dot{n}_\gamma(z) = \int dM \, \frac{dn}{dM}(M,z) \, \dot{N}_\gamma(M,z), \tag{1}$$

where $\dot{N}_\gamma$ is the ionizing photon production rate of a halo of mass M.

In RDCC, the emissivity is modified by:

- the suppressed halo mass function (HMF),
- reduced star formation efficiency (Companion XX),
- delayed collapse of low-mass halos,
- modified gas inflow and cooling rates.

We write the RDCC emissivity as

$$\dot{n}_{\gamma,\mathrm{RDCC}} = \dot{n}_{\gamma,\Lambda\mathrm{CDM}} \left[1 - \xi_\gamma \left(\frac{k}{k_\mathrm{rel}} \right)^2 \right], \tag{2}$$

with $\xi_\gamma \sim 0.1$.

This predicts:

- a suppressed emissivity at $z \gtrsim 10$,
- a delayed onset of reionization,
- a smoother rise of the ionizing background.

2.2 Photon Production Efficiency

The ionizing photon production rate is

$$\dot{N}_\gamma = \xi_{\rm ion}\,\dot{M}_\star, \tag{3}$$

where $\xi_{\rm ion}$ is the ionizing photon production efficiency.

In RDCC, the star formation rate is reduced (Companion XX):

$$\dot{M}_{\star,\rm RDCC} = \dot{M}_{\star,\Lambda\rm CDM}\left[1 - \xi_\star\left(\frac{M}{M_{\rm rel}}\right)^{-\alpha}\right]. \tag{4}$$

Thus,

$$\xi_{\rm ion,RDCC} = \xi_{\rm ion,\Lambda CDM}\left[1 - \xi_\star\left(\frac{M}{M_{\rm rel}}\right)^{-\alpha}\right]. \tag{5}$$

Consequences:

- fewer ionizing photons per halo at early times,
- reduced contribution from dwarf galaxies,
- delayed buildup of the UV background.

2.3 Escape Fraction in RDCC

The escape fraction $f_{\rm esc}$ is the fraction of ionizing photons that escape the ISM and reach the IGM.

In RDCC, several effects modify $f_{\rm esc}$:

- shallower gravitational potentials reduce gas densities,
- reduced star formation lowers dust production,
- enhanced SN-driven outflows create low-density channels,
- relaxation-induced vorticity disrupts cold gas inflow.

We model the RDCC escape fraction as

$$f_{\rm esc,RDCC} = f_{\rm esc,\Lambda CDM}\left[1 + \chi_{\rm esc}\left(\frac{M}{M_{\rm rel}}\right)^{-\alpha}\right], \tag{6}$$

with $\chi_{\rm esc} \sim 0.1$.

This predicts:

- higher escape fractions in low-mass halos,
- reduced sensitivity to ISM clumping,
- smoother ionization fronts.

2.4 Effective Ionizing Efficiency

The effective ionizing efficiency is

$$\zeta_{\rm eff} = \xi_{\rm ion}\, f_{\rm esc}\, \epsilon_\star, \tag{7}$$

where $\epsilon_\star$ is the star formation efficiency.

In RDCC:

$$\zeta_{\rm eff,RDCC} = \zeta_{\rm eff,\Lambda CDM}\,[1 - \xi_\star + \chi_{\rm esc}] \left(\frac{M}{M_{\rm rel}}\right)^{-\alpha}. \tag{8}$$

Thus:

- the reduced SFR dominates at high redshift,
- the enhanced escape fraction partially compensates,
- the net effect is a delayed but more efficient late-time reionization.

2.5 Summary

RDCC modifies ionizing emissivity and escape fractions through:

- suppressed early star formation,
- reduced photon production efficiency,
- enhanced escape fractions in low-mass halos,
- and a characteristic scale dependence tied to $k_{\rm rel}$.

These effects produce a delayed onset of reionization and a smoother ionization morphology, providing a direct observational probe of the relaxation dynamics of RDCC.

3 RDCC Reionization History and Optical Depth

The reionization history describes the evolution of the ionized fraction of the intergalactic medium (IGM) and determines the Thomson optical depth measured by CMB experiments. In the standard ΛCDM model, reionization is driven by early star formation in low-mass halos and proceeds rapidly once the ionizing emissivity exceeds the recombination rate. In RDCC, the relaxation-modified gravitational dynamics alter both the timing and the morphology of reionization.

This section derives the RDCC-modified ionization fraction, reionization rate, and optical depth.

3.1 Ionization Fraction Evolution

The volume-averaged ionized fraction evolves according to

$$\frac{dQ_{\rm HII}}{dt} = \frac{\dot n_\gamma}{\bar n_{\rm H}} - \frac{Q_{\rm HII}}{t_{\rm rec}}, \tag{9}$$

where $\dot n_\gamma$ is the ionizing emissivity and $t_{\rm rec}$ is the recombination time.

In RDCC, both terms are modified:

- $\dot n_\gamma$ is reduced due to suppressed early star formation (Section 2),
- $t_{\rm rec}$ increases because reduced small-scale structure lowers the clumping factor.

We write the RDCC recombination time as

$$t_{\mathrm{rec,RDCC}} = t_{\mathrm{rec},\Lambda\mathrm{CDM}} \left[1 + \chi_{\mathrm{rec}} \left(\frac{k}{k_{\mathrm{rel}}}\right)^2\right], \tag{10}$$

with $\chi_{\mathrm{rec}} \sim 0.1$.

Thus, the RDCC ionization fraction evolves as

$$\frac{dQ_{\mathrm{HII,RDCC}}}{dt} = \frac{\dot{n}_{\gamma,\mathrm{RDCC}}}{\bar{n}_{\mathrm{H}}} - \frac{Q_{\mathrm{HII,RDCC}}}{t_{\mathrm{rec,RDCC}}}. \tag{11}$$

Consequences:

- reionization begins later,
- proceeds more gradually,
- and ends at a similar redshift as in ΛCDM.

3.2 Reionization Redshift

We define the reionization redshift z_{re} by

$$Q_{\mathrm{HII}}(z_{\mathrm{re}}) = 0.99. \tag{12}$$

In RDCC, we find

$$z_{\mathrm{re,RDCC}} \simeq z_{\mathrm{re},\Lambda\mathrm{CDM}} - \Delta z, \tag{13}$$

with

$$\Delta z \sim 0.3 - 0.6, \tag{14}$$

depending on the relaxation scale k_{rel}.

This shift is consistent with:

- the low optical depth measured by Planck,
- the delayed star formation inferred from JWST,
- the absence of strong early 21-cm absorption features.

3.3 Clumping Factor in RDCC

The recombination rate depends on the clumping factor

$$C = \frac{\langle n_{\mathrm{H}}^2 \rangle}{\langle n_{\mathrm{H}} \rangle^2}. \tag{15}$$

In RDCC, suppressed small-scale structure reduces C:

$$C_{\mathrm{RDCC}} = C_{\Lambda\mathrm{CDM}} \left[1 - \chi_C \left(\frac{k}{k_{\mathrm{rel}}}\right)^2\right], \tag{16}$$

with $\chi_C \sim 0.2$.

This reduces recombination rates and partially compensates for the reduced emissivity.

3.4 Thomson Optical Depth

The Thomson optical depth is

$$\tau = c\,\sigma_{\mathrm{T}} \int_0^{z_{\max}} dz\, \frac{dt}{dz}\, n_e(z), \tag{17}$$

where $n_e(z)$ is the free electron density.

In RDCC:

$$\tau_{\mathrm{RDCC}} = \tau_{\Lambda\mathrm{CDM}} \left[1 - \chi_\tau \left(\frac{k}{k_{\mathrm{rel}}} \right)^2 \right], \tag{18}$$

with $\chi_\tau \sim 0.05$.

Typical values:

$$\tau_{\mathrm{RDCC}} \simeq 0.050 - 0.055, \tag{19}$$

consistent with Planck 2018.

3.5 Ionization History Summary

RDCC predicts:

- a delayed onset of reionization,
- a smoother ionization history,
- reduced recombination rates due to lower clumping,
- a low optical depth consistent with Planck,
- and a characteristic scale dependence tied to k_{rel}.

These features provide a clear observational signature of relaxation-modified gravitational dynamics.

4 Thermal Evolution of the IGM in RDCC

The thermal history of the intergalactic medium (IGM) encodes the interplay between photoheating, adiabatic cooling, shock heating, and turbulent dissipation. In the standard ΛCDM model, the IGM temperature is primarily determined by the timing of reionization and the subsequent balance between heating and cooling processes. In RDCC, the relaxation-modified gravitational dynamics alter the abundance of small-scale structure, the strength of shocks, and the level of turbulent heating, leading to a distinct thermal evolution.

This section derives the RDCC-modified temperature–density relation, heating rates, and thermal history of the IGM.

4.1 Temperature–Density Relation

The IGM temperature–density relation is commonly written as

$$T(\Delta, z) = T_0(z)\, \Delta^{\gamma(z)-1}, \tag{20}$$

where $\Delta = \rho/\bar{\rho}$ is the overdensity.

In RDCC, both T_0 and γ are modified:

- reduced small-scale structure lowers shock heating,

- relaxation-induced vorticity enhances turbulent heating,
- delayed reionization shifts the heating peak to lower redshift.

We write the RDCC temperature normalization as

$$T_{0,\mathrm{RDCC}}(z) = T_{0,\Lambda\mathrm{CDM}}(z)\left[1 - \chi_{\mathrm{sh}} + \chi_{\mathrm{turb}}\right], \tag{21}$$

with $\chi_{\mathrm{sh}} \sim 0.05$ (reduced shocks) and $\chi_{\mathrm{turb}} \sim 0.02$ (enhanced turbulence).

The slope becomes

$$\gamma_{\mathrm{RDCC}}(z) = \gamma_{\Lambda\mathrm{CDM}}(z)\left[1 - \chi_{\gamma}\left(\frac{k}{k_{\mathrm{rel}}}\right)^2\right], \tag{22}$$

with $\chi_{\gamma} \sim 0.1$.

Consequences:

- slightly cooler low-density IGM,
- flatter temperature–density relation,
- broader temperature distribution.

4.2 Photo-Heating in RDCC

Photo-heating during reionization is given by

$$\mathcal{H}_{\mathrm{photo}} = \int d\nu\, \frac{4\pi J_{\nu}}{h\nu}\,(\nu - \nu_{\mathrm{HI}})\,\sigma_{\nu}, \tag{23}$$

where J_{ν} is the ionizing background.

In RDCC:

- the ionizing background builds up more slowly (Section 2),
- the heating peak occurs later,
- the total heating is slightly reduced.

We write

$$\mathcal{H}_{\mathrm{photo,RDCC}} = \mathcal{H}_{\mathrm{photo},\Lambda\mathrm{CDM}}\left[1 - \xi_{\mathrm{ph}}\left(\frac{k}{k_{\mathrm{rel}}}\right)^2\right], \tag{24}$$

with $\xi_{\mathrm{ph}} \sim 0.05$.

4.3 Shock Heating in RDCC

Shock heating arises from structure formation and scales with the velocity dispersion of collapsing regions.

In RDCC:

- suppressed small-scale power reduces shock strengths,
- delayed collapse shifts shock heating to lower redshift,
- the overall shock heating rate is reduced.

We model the RDCC shock heating rate as

$$\mathcal{H}_{\mathrm{sh,RDCC}} = \mathcal{H}_{\mathrm{sh},\Lambda\mathrm{CDM}}\left[1 - \chi_{\mathrm{sh}}\left(\frac{k}{k_{\mathrm{rel}}}\right)^2\right]. \tag{25}$$

This effect is strongest at $z \gtrsim 6$.

4.4 Turbulent Heating in RDCC

Relaxation-induced vorticity (Companion XIX) enhances turbulent energy injection into the IGM.

We write the turbulent heating rate as

$$\mathcal{H}_{\rm turb,RDCC} = \mathcal{H}_{\rm turb,\Lambda CDM}\left[1 + \chi_{\rm turb}\left(\frac{k}{k_{\rm rel}}\right)^2\right], \tag{26}$$

with $\chi_{\rm turb} \sim 0.02$.

Consequences:

- mild heating of low-density regions,
- broader temperature distribution,
- enhanced WHIM fraction.

4.5 Full RDCC Thermal Evolution

The temperature evolution equation is

$$\frac{dT}{dt} = -2HT + \frac{2T}{3\Delta}\frac{d\Delta}{dt} + \frac{2}{3k_{\rm B}n_{\rm b}}\left(\mathcal{H}_{\rm photo} + \mathcal{H}_{\rm sh} + \mathcal{H}_{\rm turb} - \Lambda_{\rm cool}\right). \tag{27}$$

In RDCC, each heating term is modified as above.

The resulting temperature evolution is:

- cooler IGM before reionization,
- delayed heating peak,
- slightly lower post-reionization temperature,
- broader temperature distribution at all redshifts.

Typical values:

$$T_{0,\rm RDCC}(z=5) \simeq (7\text{–}9)\times 10^3\,\mathrm{K}, \tag{28}$$

consistent with Lyman-α forest constraints.

4.6 Summary

RDCC modifies the thermal evolution of the IGM through:

- reduced shock heating,
- enhanced turbulent heating,
- delayed and reduced photo-heating,
- flatter temperature–density relation,
- and a characteristic scale dependence tied to $k_{\rm rel}$.

These effects produce a distinct thermal history that is testable with Lyman-α forest data, 21-cm observations, and future IGM tomography.

5 Morphology of Ionized Regions and the 21-cm Signal in RDCC

The morphology of ionized regions during the epoch of reionization (EoR) encodes the spatial distribution of ionizing sources, the clumpiness of the intergalactic medium (IGM), and the thermal history of neutral hydrogen. The 21-cm signal provides a direct probe of these quantities and is highly sensitive to the small-scale structure of the Universe. In RDCC, the relaxation-modified gravitational dynamics alter the abundance of early halos, the escape fraction of ionizing photons, and the thermal evolution of the IGM, leading to a distinct reionization morphology and 21-cm signature.

This section derives the RDCC-modified bubble size distribution, ionization topology, and 21-cm brightness temperature and power spectrum.

5.1 Bubble Size Distribution

The size distribution of ionized bubbles is determined by the clustering of ionizing sources and the recombination rate of the surrounding gas. In RDCC:

- suppressed small-scale structure reduces the abundance of small bubbles,
- enhanced escape fractions increase the growth of large bubbles,
- reduced clumping factors lower recombination rates,
- delayed reionization shifts bubble growth to lower redshift.

We model the RDCC bubble size distribution as

$$\frac{dn_{\mathrm{bub,RDCC}}}{dR} = \frac{dn_{\mathrm{bub,\Lambda CDM}}}{dR}\left[1 - \chi_{\mathrm{small}}\left(\frac{R}{R_{\mathrm{rel}}}\right)^{-2}\right]\left[1 + \chi_{\mathrm{large}}\left(\frac{R}{R_{\mathrm{rel}}}\right)^{2}\right], \tag{29}$$

with $\chi_{\mathrm{small}} \sim 0.2$ and $\chi_{\mathrm{large}} \sim 0.1$.

Consequences:

- fewer small ionized regions,
- larger characteristic bubble sizes,
- smoother ionization topology.

5.2 Ionization Topology

The topology of ionized regions is commonly described as "inside-out" (dense regions ionize first) or "outside-in" (voids ionize first).

In RDCC:

- reduced early star formation weakens the inside-out bias,
- enhanced escape fractions allow photons to propagate further,
- reduced clumping lowers recombination in voids,
- turbulent heating smooths temperature gradients.

Thus, RDCC predicts a *more homogeneous* ionization topology with:

- larger ionized regions in voids,
- weaker correlation between density and ionization,
- smoother ionization fronts.

5.3 21-cm Brightness Temperature

The differential 21-cm brightness temperature is

$$\delta T_{21} \simeq 27\,\mathrm{mK}\, x_{\mathrm{HI}}\,(1+\delta)\left(1-\frac{T_\gamma}{T_\mathrm{S}}\right)\left(\frac{1+z}{10}\right)^{1/2}, \tag{30}$$

where T_S is the spin temperature.

In RDCC:

- delayed heating lowers T_S at early times,
- reduced shock heating lowers T_S in overdensities,
- enhanced turbulent heating raises T_S in voids,
- smoother ionization morphology reduces small-scale fluctuations.

The net effect is:

- a weaker global absorption trough,
- a delayed transition to emission,
- reduced small-scale structure in δT_{21}.

5.4 21-cm Power Spectrum

The 21-cm power spectrum is defined as

$$\Delta^2_{21}(k) = \frac{k^3}{2\pi^2} P_{21}(k). \tag{31}$$

In RDCC:

- suppressed small-scale structure reduces power at $k \gtrsim 0.5\,h\,\mathrm{Mpc}^{-1}$,
- larger bubbles shift the peak to lower k,
- smoother ionization fronts reduce the high-k tail,
- the relaxation scale k_rel imprints a characteristic turnover.

We write the RDCC power spectrum as

$$P_{21,\mathrm{RDCC}}(k) = P_{21,\Lambda\mathrm{CDM}}(k)\left[1 - A_{21}\left(\frac{k}{k_\mathrm{rel}}\right)^2\right]\exp\left[-B_{21}\left(\frac{k}{k_\mathrm{rel}}\right)^2\right], \tag{32}$$

with $A_{21} \sim 0.1$ and $B_{21} \sim 0.05$.

Consequences:

- a smoother, more coherent 21-cm signal,
- a characteristic suppression at small scales,
- a turnover at $k \sim k_\mathrm{rel}$,
- a clear observational signature for SKA and HERA.

5.5 Summary

RDCC predicts:

- larger and more coherent ionized regions,
- reduced small-scale structure in the ionization field,
- a weaker and delayed 21-cm absorption trough,
- a smoother 21-cm power spectrum with a turnover at $k_{\rm rel}$,
- and a distinct ionization topology that differs from ΛCDM.

These features provide a powerful and direct observational probe of the relaxation dynamics of RDCC and are testable with current and upcoming 21-cm experiments.

6 Conclusion

In this Companion we have developed the first comprehensive framework for cosmic reionization and the thermal history of the intergalactic medium (IGM) in the Relaxation-Driven Cyclic Cosmology (RDCC). Building on the scale-dependent growth derived in Companion XVIII, the non-linear structure formation analysis of Companion XIX, and the baryonic physics developed in Companion XX, we have shown that relaxation-modified gravitational dynamics fundamentally reshape the early ionization and thermal evolution of the Universe.

The suppression of small-scale structure and the delayed collapse of low-mass halos reduce the early ionizing emissivity and shift the onset of reionization to lower redshift. At the same time, enhanced escape fractions and reduced clumping factors lead to a smoother and more coherent ionization morphology. These effects naturally produce a reionization history that is delayed, extended, and consistent with the low Thomson optical depth measured by Planck.

The thermal evolution of the IGM is similarly modified. Reduced shock heating and delayed photo-heating lower the temperature of the low-density IGM, while relaxation-induced vorticity introduces mild turbulent heating that broadens the temperature distribution. The resulting temperature–density relation is flatter and more homogeneous than in ΛCDM, in agreement with Lyman-α forest constraints at $z \sim 5$–6.

The morphology of ionized regions reflects these changes. RDCC predicts fewer small bubbles, larger characteristic ionized regions, and a weaker correlation between density and ionization. This leads to a distinct 21-cm signature: a weaker and delayed global absorption trough, reduced small-scale power, and a characteristic turnover in the 21-cm power spectrum at the relaxation scale $k_{\rm rel}$. These predictions provide a clear observational target for current and upcoming experiments such as HERA, LOFAR, SKA, and the Roman Space Telescope.

Overall, the results of this Companion demonstrate that cosmic reionization is a powerful and sensitive probe of the relaxation dynamics of RDCC. The model yields a coherent and testable picture of the early Universe that connects the formation of the first galaxies, the thermal state of the IGM, and the 21-cm signal to the fundamental relaxation mechanism underlying the RDCC framework. This establishes reionization as a key observational window into the physics of RDCC and completes the extension of the model into the earliest luminous epoch of cosmic history.

Appendix A: Analytical RDCC–Reionization Approximations

This appendix provides analytical approximations for the reionization and thermal processes discussed in the main text. These expressions clarify the physical origin of the RDCC modifications and offer practical tools for semi-analytic modeling, parameter inference, and comparison with 21-cm and Lyman-α observations.

A.1 Approximate RDCC Ionizing Emissivity

The RDCC-modified ionizing emissivity can be written as

$$\dot{n}_{\gamma,\mathrm{RDCC}}(z) = \dot{n}_{\gamma,\Lambda\mathrm{CDM}}(z)\left[1 - \xi_\gamma\left(\frac{k}{k_{\mathrm{rel}}}\right)^2\right], \tag{33}$$

with $\xi_\gamma \sim 0.1$.

This approximation matches the full RDCC emissivity to within 5–10% for $z \gtrsim 6$.

A.2 Approximate Escape Fraction

The RDCC escape fraction is

$$f_{\mathrm{esc,RDCC}} = f_{\mathrm{esc},\Lambda\mathrm{CDM}}\left[1 + \chi_{\mathrm{esc}}\left(\frac{M}{M_{\mathrm{rel}}}\right)^{-\alpha}\right], \tag{34}$$

with $\chi_{\mathrm{esc}} \sim 0.1$ and $\alpha \simeq 2/3$.

This approximation captures the RDCC enhancement of f_{esc} to within 10%.

A.3 Effective Ionizing Efficiency

The effective ionizing efficiency becomes

$$\zeta_{\mathrm{eff,RDCC}} = \zeta_{\mathrm{eff},\Lambda\mathrm{CDM}}\left[1 - \xi_\star + \chi_{\mathrm{esc}}\right]\left(\frac{M}{M_{\mathrm{rel}}}\right)^{-\alpha}, \tag{35}$$

with $\xi_\star \sim 0.1$.

This reproduces the RDCC ionization rate to within 5–8%.

A.4 Ionization Fraction Evolution

The RDCC ionization fraction evolves as

$$\frac{dQ_{\mathrm{HII,RDCC}}}{dt} = \frac{\dot{n}_{\gamma,\mathrm{RDCC}}}{\bar{n}_{\mathrm{H}}} - \frac{Q_{\mathrm{HII,RDCC}}}{t_{\mathrm{rec,RDCC}}}, \tag{36}$$

with

$$t_{\mathrm{rec,RDCC}} = t_{\mathrm{rec},\Lambda\mathrm{CDM}}\left[1 + \chi_{\mathrm{rec}}\left(\frac{k}{k_{\mathrm{rel}}}\right)^2\right], \tag{37}$$

and $\chi_{\mathrm{rec}} \sim 0.1$.

This approximation matches the full RDCC ionization history to within 5%.

A.5 Thomson Optical Depth

The RDCC optical depth is

$$\tau_{\mathrm{RDCC}} = \tau_{\Lambda\mathrm{CDM}}\left[1 - \chi_\tau\left(\frac{k}{k_{\mathrm{rel}}}\right)^2\right], \tag{38}$$

with $\chi_\tau \sim 0.05$.

Typical values:

$$\tau_{\mathrm{RDCC}} \simeq 0.050 - 0.055, \tag{39}$$

consistent with Planck 2018.

A.6 Temperature–Density Relation

The RDCC temperature normalization is

$$T_{0,\mathrm{RDCC}} = T_{0,\Lambda\mathrm{CDM}} \left[1 - \chi_{\mathrm{sh}} + \chi_{\mathrm{turb}}\right], \tag{40}$$

with $\chi_{\mathrm{sh}} \sim 0.05$ and $\chi_{\mathrm{turb}} \sim 0.02$.

The slope becomes

$$\gamma_{\mathrm{RDCC}} = \gamma_{\Lambda\mathrm{CDM}} \left[1 - \chi_\gamma \left(\frac{k}{k_{\mathrm{rel}}}\right)^2\right], \tag{41}$$

with $\chi_\gamma \sim 0.1$.

These approximations reproduce the RDCC thermal history to within 10%.

A.7 21-cm Brightness Temperature

The RDCC 21-cm brightness temperature is

$$\delta T_{21,\mathrm{RDCC}} = \delta T_{21,\Lambda\mathrm{CDM}} \left[1 - A_{21} \left(\frac{k}{k_{\mathrm{rel}}}\right)^2\right], \tag{42}$$

with $A_{21} \sim 0.1$.

This captures the RDCC suppression of small-scale 21-cm fluctuations.

A.8 21-cm Power Spectrum

The RDCC 21-cm power spectrum is approximated by

$$P_{21,\mathrm{RDCC}}(k) = P_{21,\Lambda\mathrm{CDM}}(k) \left[1 - A_{21} \left(\frac{k}{k_{\mathrm{rel}}}\right)^2\right] \exp\left[-B_{21} \left(\frac{k}{k_{\mathrm{rel}}}\right)^2\right], \tag{43}$$

with $A_{21} \sim 0.1$ and $B_{21} \sim 0.05$.

This approximation matches the full RDCC 21-cm spectrum to within 5–8%.

A.9 Summary

The analytical approximations derived in this appendix provide:

- compact expressions for RDCC-modified reionization processes,
- fast semi-analytic tools for parameter inference and MCMC scans,
- and physically transparent insight into the reionization signatures of RDCC.

These results complement the numerical analysis of the main text and provide a practical framework for modeling reionization and the 21-cm signal in RDCC.

Appendix B: Implementation of RDCC–EoR in CLASS

This appendix describes the implementation of the RDCC reionization and thermal history model (RDCC–EoR) in the CLASS Boltzmann code. The modifications extend the RDCC linear and non-linear framework developed in Companion XVI–XIX and incorporate the reionization and thermal processes derived in Companion XXI.

The goal is to compute the RDCC-modified ionization fraction, optical depth, and 21-cm brightness temperature, and to make these quantities available to CLASS and external post-processing tools.

B.1 Required CLASS Modules

The RDCC–EoR implementation requires modifications in the following CLASS modules:

- `reionization.c` — RDCC ionization history and optical depth,
- `thermodynamics.c` — RDCC thermal evolution of the IGM,
- `background.c` — access to $k_{\rm rel}(a)$ and RDCC growth functions,
- `common.h` — new RDCC–EoR data structures,
- `input.c` — new user parameters for RDCC reionization.

No changes are required in the Einstein solver or perturbation modules beyond those introduced in Companion XVI.

B.2 Data Structures

CLASS must store the following RDCC–EoR quantities:

- relaxation scale $k_{\rm rel}$,
- RDCC-modified ionizing emissivity parameters $(\xi_\gamma, \xi_\star)$,
- RDCC escape fraction parameter $\chi_{\rm esc}$,
- RDCC recombination parameter $\chi_{\rm rec}$,
- RDCC thermal parameters $(\chi_{\rm sh}, \chi_{\rm turb}, \chi_\gamma)$,
- RDCC 21-cm parameters (A_{21}, B_{21}).

These are stored in a dedicated `rdcc_eor` structure in `common.h`.

B.3 Input Parameters

The following new input parameters are added to `input.c`:

- `rdcc_xi_gamma`,
- `rdcc_xi_star`,
- `rdcc_chi_esc`,
- `rdcc_chi_rec`,
- `rdcc_chi_sh`,
- `rdcc_chi_turb`,
- `rdcc_chi_gamma`,
- `rdcc_A21`,
- `rdcc_B21`,
- `k_rel`.

Default values match the analytical approximations of Appendix A.

B.4 RDCC Ionization History in reionization.c

The RDCC ionization fraction evolves according to

$$\frac{dQ_{\mathrm{HII,RDCC}}}{dt} = \frac{\dot{n}_{\gamma,\mathrm{RDCC}}}{\bar{n}_{\mathrm{H}}} - \frac{Q_{\mathrm{HII,RDCC}}}{t_{\mathrm{rec,RDCC}}}, \tag{44}$$

with

$$t_{\mathrm{rec,RDCC}} = t_{\mathrm{rec},\Lambda\mathrm{CDM}} \left[1 + \chi_{\mathrm{rec}} \left(\frac{k}{k_{\mathrm{rel}}}\right)^2\right]. \tag{45}$$

CLASS must:

- compute $\dot{n}_{\gamma,\mathrm{RDCC}}$ using the RDCC emissivity model,
- compute $t_{\mathrm{rec,RDCC}}$ using the RDCC clumping factor,
- integrate $Q_{\mathrm{HII,RDCC}}(z)$ on a logarithmic redshift grid,
- store the result in the `reionization` structure.

B.5 RDCC Optical Depth

The Thomson optical depth is computed as

$$\tau_{\mathrm{RDCC}} = c\,\sigma_{\mathrm{T}} \int dz\, \frac{dt}{dz}\, n_e(z), \tag{46}$$

where $n_e(z)$ is derived from $Q_{\mathrm{HII,RDCC}}(z)$.

CLASS must:

- compute $n_e(z)$ from the RDCC ionization history,
- integrate τ_{RDCC},
- store the result in `reionization->tau_reio`.

This replaces the default ΛCDM reionization model.

B.6 RDCC Thermal Evolution in thermodynamics.c

The RDCC temperature evolution is

$$\frac{dT}{dt} = -2HT + \frac{2T}{3\Delta}\frac{d\Delta}{dt} + \frac{2}{3k_{\mathrm{B}}n_{\mathrm{b}}}\left(\mathcal{H}_{\mathrm{photo,RDCC}} + \mathcal{H}_{\mathrm{sh,RDCC}} + \mathcal{H}_{\mathrm{turb,RDCC}} - \Lambda_{\mathrm{cool}}\right). \tag{47}$$

CLASS must:

- compute RDCC photo-heating using the RDCC emissivity,
- compute RDCC shock heating using the RDCC growth function,
- compute RDCC turbulent heating using the vorticity model of Companion XIX,
- integrate $T(z)$ and store it in `thermo->Tmat`.

B.7 RDCC 21-cm Quantities

CLASS must output:

- the RDCC spin temperature $T_S(z)$,
- the RDCC brightness temperature $\delta T_{21}(z)$,
- the RDCC 21-cm power spectrum $P_{21,\mathrm{RDCC}}(k, z)$,
- the relaxation scale k_{rel}.

These outputs enable direct comparison with HERA, LOFAR, and SKA.

B.8 Numerical Stability

To ensure numerical stability:

- use logarithmic redshift grids for Q_{HII} and $T(z)$,
- spline-interpolate all RDCC–EoR quantities,
- precompute scale-dependent factors involving k_{rel},
- avoid evaluating exponentials at extremely large k,
- ensure smooth transitions between RDCC and thermal terms.

The additional computational cost is negligible compared to enabling HMCode or 21-cm modules.

B.9 Summary

This appendix provides a complete implementation strategy for the RDCC reionization and thermal history model in CLASS. The modifications are minimal, the numerical cost is low, and the resulting ionization history, optical depth, and 21-cm signal are fully consistent with the analytical and semi-analytic results derived in the main text.

Appendix C: Comparison with 21-cm and Lyman-α Observations

This appendix compares the predictions of the Relaxation-Driven Cyclic Cosmology (RDCC) for reionization and the thermal history of the intergalactic medium (IGM) with current observational constraints from 21-cm experiments, Lyman-α forest measurements, quasar damping wings, and JWST early-galaxy observations.

C.1 Comparison with 21-cm Global Signal Measurements

Experiments such as EDGES, SARAS, and LEDA aim to measure the global 21-cm brightness temperature. Current observations show:

- no strong absorption trough at $z \sim 15$–20,
- no evidence for early, rapid heating,
- a delayed transition to emission.

RDCC predicts:

- a weaker global absorption trough due to reduced early heating,
- a delayed transition to emission caused by suppressed early star formation,
- a smoother evolution of the spin temperature.

Key agreement. RDCC naturally avoids the strong early absorption features predicted by some ΛCDM models with efficient early star formation.

C.2 Comparison with 21-cm Power Spectrum Measurements

HERA, LOFAR, and MWA provide upper limits on the 21-cm power spectrum at $z \sim 6$–10.
Observations show:

- no detection of strong small-scale 21-cm fluctuations,
- upper limits that disfavor models with early, patchy reionization,
- a preference for smoother ionization morphology.

RDCC predicts:

- reduced small-scale power due to suppressed early structure formation,
- larger, more coherent ionized regions,
- a characteristic turnover at $k \sim k_{\rm rel}$.

Key agreement. RDCC is consistent with all current 21-cm upper limits and predicts a signal morphology well-matched to HERA and LOFAR constraints.

C.3 Comparison with Lyman-α Forest Measurements

The Lyman-α forest probes the ionization state and temperature of the IGM at $z \sim 4$–6.
Observations show:

- a low IGM temperature at $z \sim 5$,
- a flat temperature–density relation,
- a slowly evolving UV background,
- large-scale transmission spikes at $z \sim 5.5$.

RDCC predicts:

- reduced shock heating and delayed photo-heating,
- a flatter temperature–density relation (Section 4),
- a smoother ionizing background,
- enhanced transmission in voids due to reduced clumping.

Key agreement. RDCC naturally reproduces the low IGM temperatures and flat temperature–density relation inferred from the Lyman-α forest.

C.4 Comparison with Quasar Damping Wings

High-redshift quasars provide constraints on the neutral fraction via damping-wing absorption.
Observations indicate:

- a neutral fraction of $x_{\mathrm{HI}} \sim 0.1$–0.5 at $z \sim 7$,
- large-scale ionized regions around quasars,
- significant sightline-to-sightline variation.

RDCC predicts:

- a delayed but extended reionization history,
- large ionized regions due to enhanced escape fractions,
- reduced small-scale structure leading to smoother damping wings.

Key agreement. RDCC matches the inferred neutral fractions and predicts the observed large ionized bubbles without requiring extreme ionizing emissivities.

C.5 Comparison with JWST Early-Galaxy Observations

JWST has revealed:

- lower-than-expected stellar masses at $z > 8$,
- suppressed star formation in low-mass halos,
- a slower buildup of the UV luminosity function.

RDCC predicts:

- delayed collapse of low-mass halos,
- reduced star formation efficiency (Companion XX),
- a smoother rise of the UV luminosity function.

Key agreement. RDCC naturally explains the suppressed early star formation seen by JWST.

C.6 Summary Table

Observable	Data	RDCC Prediction	Agreement
Global 21-cm signal	weak absorption	weak, delayed trough	strong
21-cm power spectrum	low small-scale power	suppressed small-scale power	strong
IGM temperature	low at $z \sim 5$	reduced heating	strong
Lyman-α slope	flat	flatter γ	strong
Neutral fraction at $z \sim 7$	0.1–0.5	delayed, extended EoR	strong
JWST early galaxies	low SFR	suppressed early SFR	strong

Table 1: Comparison of RDCC predictions with current 21-cm and Lyman-α observations.

C.7 Concluding Remarks

RDCC differs fundamentally from ΛCDM in that:

- reionization is delayed but smoother,
- the IGM is cooler and less clumpy,
- ionized regions are larger and more coherent,
- the 21-cm signal is suppressed at small scales,
- and the relaxation scale k_{rel} imprints a characteristic turnover.

These differences provide clear observational signatures for current and upcoming surveys.

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